Recording

EN 580.133
siRNA as Therapeutics (Part II)

Elaine Zhelan Chen

Jan 13, 2025

Created by Elaine Zhelan Chen

Recap

- siRNA-mediated gene silencing mechanism
- Key challenges in siRNA-drug design
  - Off-target effects – computational design
  - Instability – chemical modifications
  - Innate immunity – chemical modifications
- siRNA targeted delivery strategy
- miRNA biogenesis
- miRNA vs siRNA

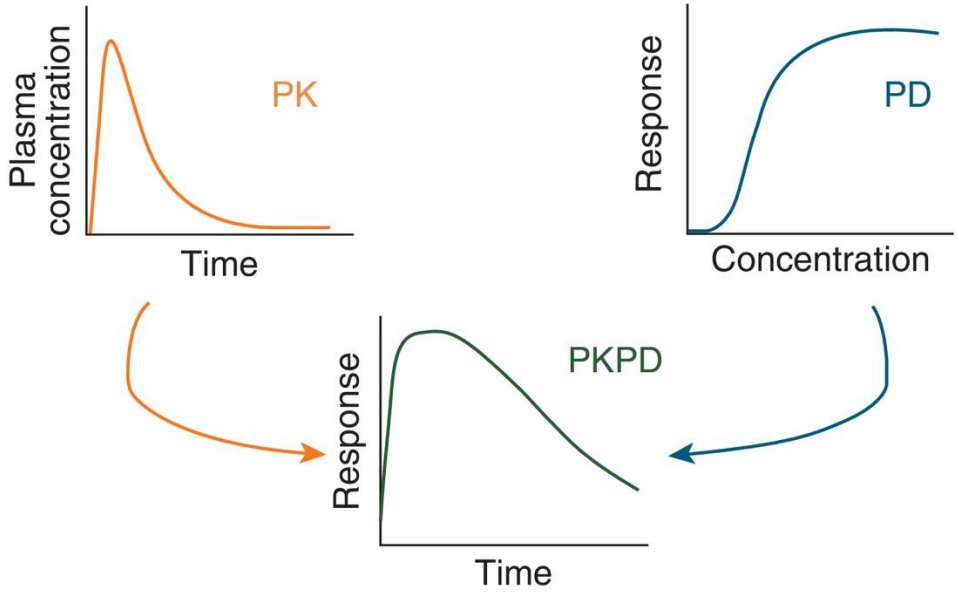
Pharmacology terms

PK and PD

- Pharmacokinetic (PK)
  - PK is the analysis of how the body affects a drug
  - PK is defined by how a compound is absorbed, distributed, metabolized, and excreted
- Pharmacodynamic (PD)
  - PD is the analysis of how a drug affects the body
  - PD is the measure of a compound's ability to interact with its intended target leading to a biologic effect

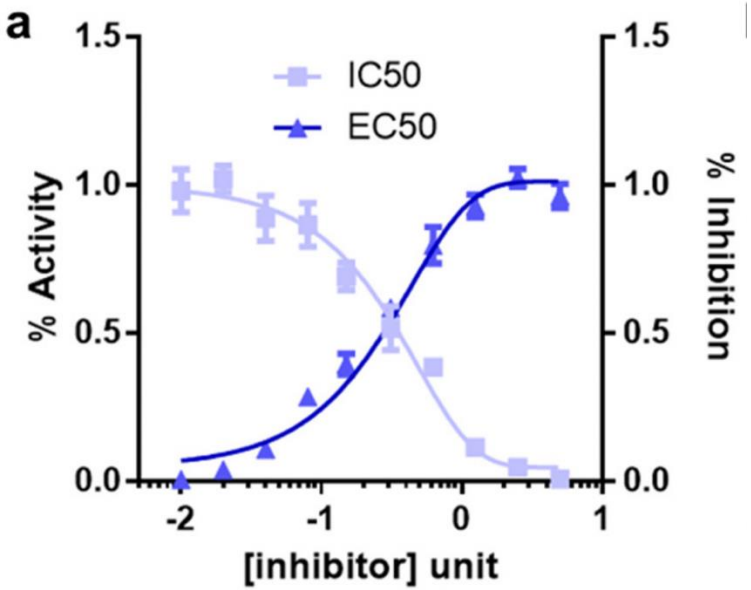
PK and PD

- PKPD allows scientists to predict the pharmacodynamic response to certain drugs which provides useful information for understanding drug action and determining dosage regimen



EC50 and IC50

- Half-maximal effective concentration (EC50)
- Half-maximal inhibitory concentration (IC50)
- A measure of a drug's efficacy and potency
- Measures the concentration of a drug or compound required to inhibit or increase the activity of a particular biological or biochemical process by 50%
- The lower the EC50 or IC50 value, the more potent the drug



Therapeutic index (TI)

- A ratio that compares the concentration at which a drug becomes toxic and the concentration at which the drug is effective

$$\text{Therapeutic Index} = \frac{LD_{50}}{ED_{50}}$$

LD_{50} = median lethal dose

ED_{50} = median effective dose

Figure 1A. Dose-Response Curves for Wide Therapeutic Index Medications

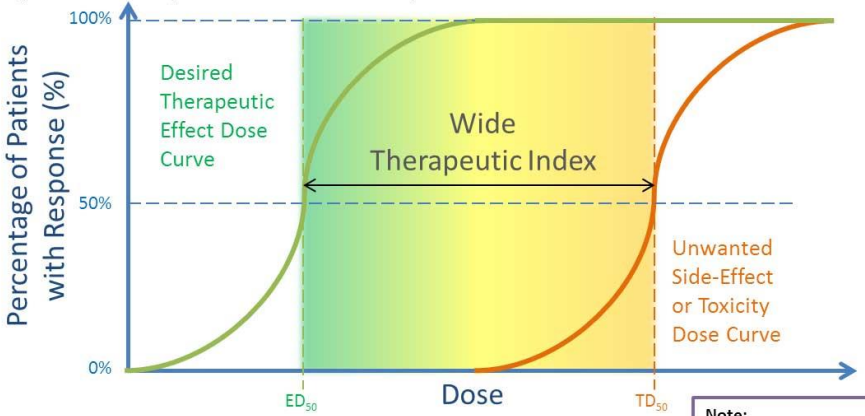
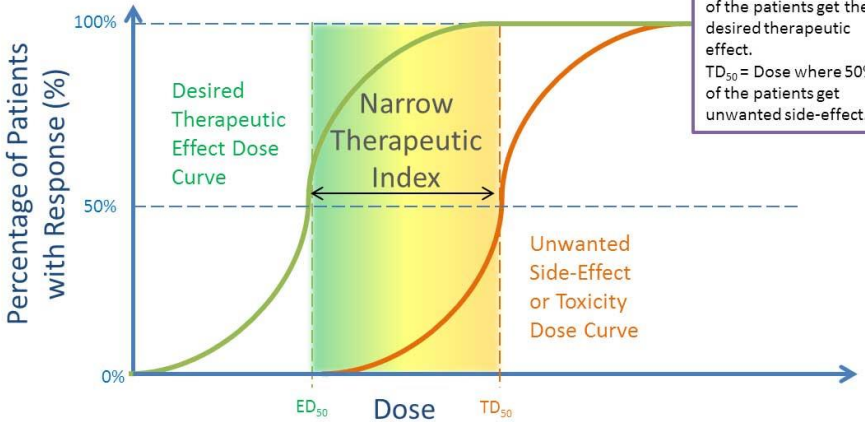


Figure 1B. Dose-Response Curves for Narrow Therapeutic Index Medications



Note:
ED_{50} = Dose where 50% of the patients get the desired therapeutic effect.
TD_{50} = Dose where 50% of the patients get unwanted side-effect.

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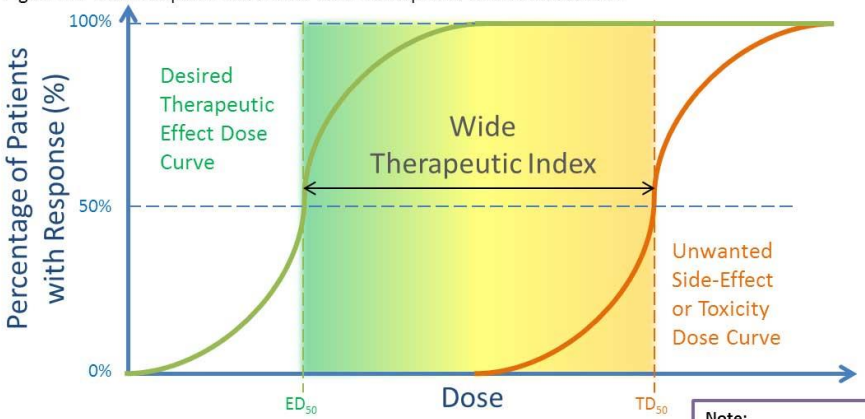
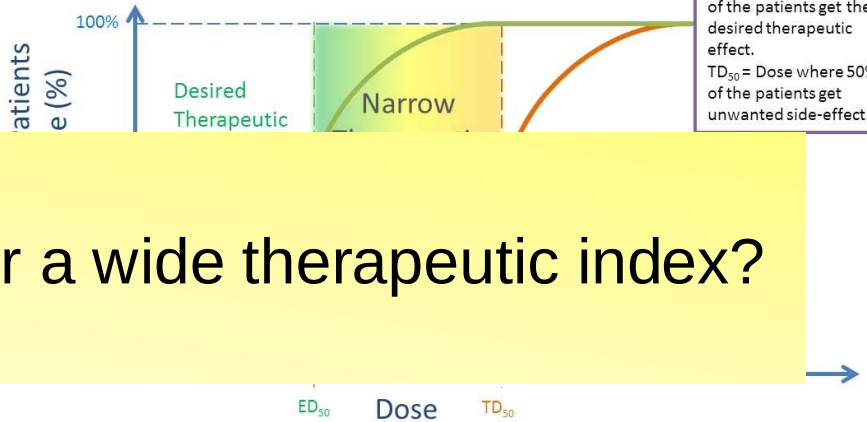


Figure 1B. Dose-Response Curves for Narrow Therapeutic Index Medications



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Which is better,
a narrow therapeutic index or a wide therapeutic index?
And why is that?

Therapeutic index (TI)

- A ratio that compares the concentration at which a drug becomes toxic and the concentration at which the drug is effective

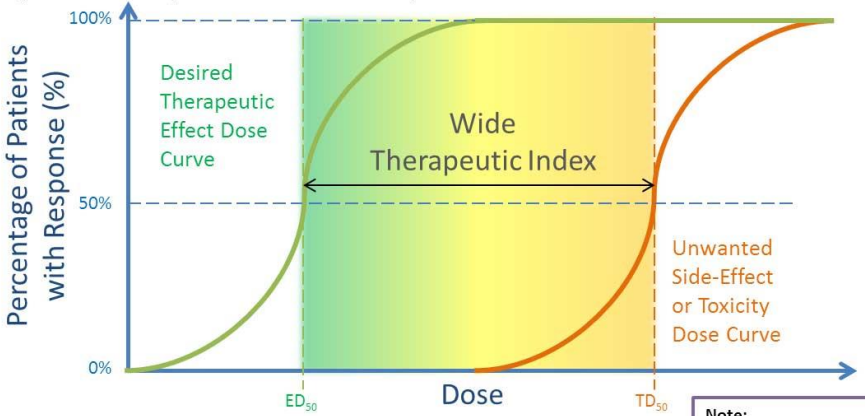
$$\text{Therapeutic Index} = \frac{LD_{50}}{ED_{50}}$$

LD_{50} = median lethal dose

ED_{50} = median effective dose

- The larger the therapeutic index (TI), the safer the drug is
- If the TI is small, the drug must be dosed carefully and the person receiving the drug should be monitored closely for any signs of drug toxicity

Figure 1A. Dose-Response Curves for Wide Therapeutic Index Medications



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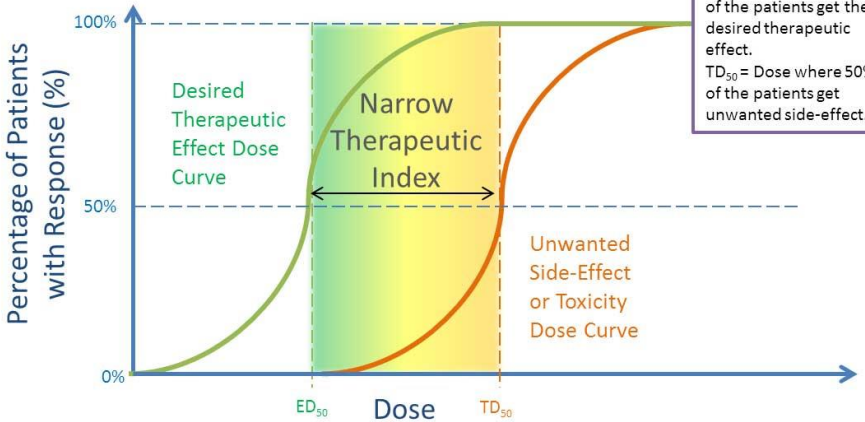

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Approved siRNA drug

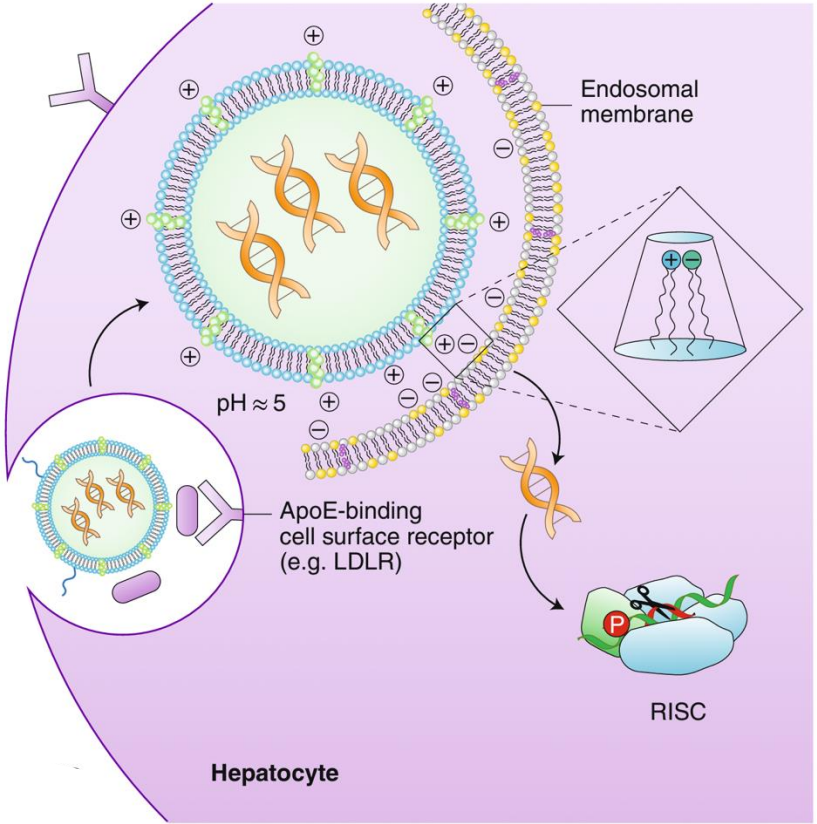
Alnylam

- “al nilam” means string of pearls in Arabic
- Founded in 2002
- All 6 siRNA therapies approved worldwide were developed or co-developed by Alnylam

Drug	Brand Name	FDA Approval	Company	Delivery System	Indication
Patisiran	Onpattro	2018	Alnylam	LNP-siRNA	Hereditary transthyretin-mediated amyloidosis
Givosiran	Givlaari	2019	Alnylam	GalNAc-siRNA	Acute hepatic porphyria
Lumasiran	Oxlumo	2020	Alnylam	GalNAc-siRNA	Primary hyperoxaluria type 1
Inclisiran	Leqvio	2021	Novartis	GalNAc-siRNA	Primary hypercholesterolemia
Vutrisiran	Amvuttra	2022	Alnylam	GalNAc-siRNA	Hereditary transthyretin-mediated amyloidosis
Nedosiran	Rivfloza	2023	Novo Nordisk	GalXC™ RNAi platform	Primary hyperoxaluria type 1

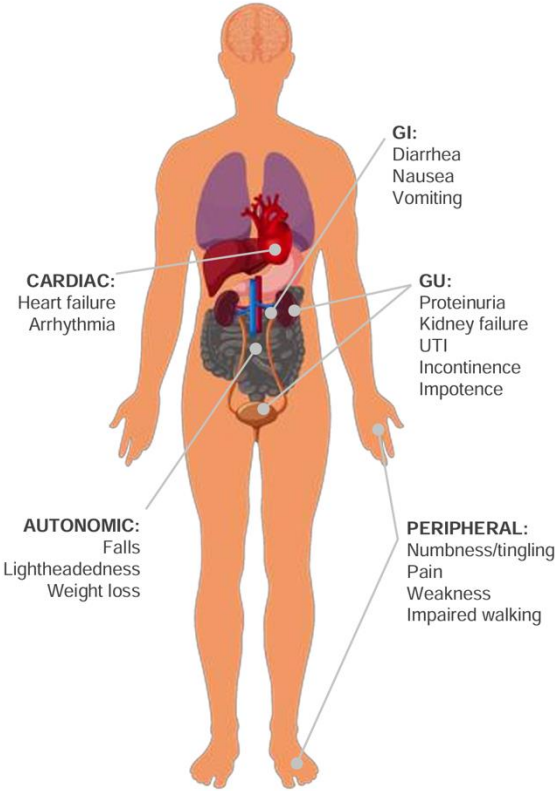
Onpattro (patisiran) – the 1st siRNA-drug

- LNP-packaged siRNA
- Approved in 2018
- For treating polyneuropathies resulting from the hereditary transthyretin-mediated amyloidosis (hATTR)



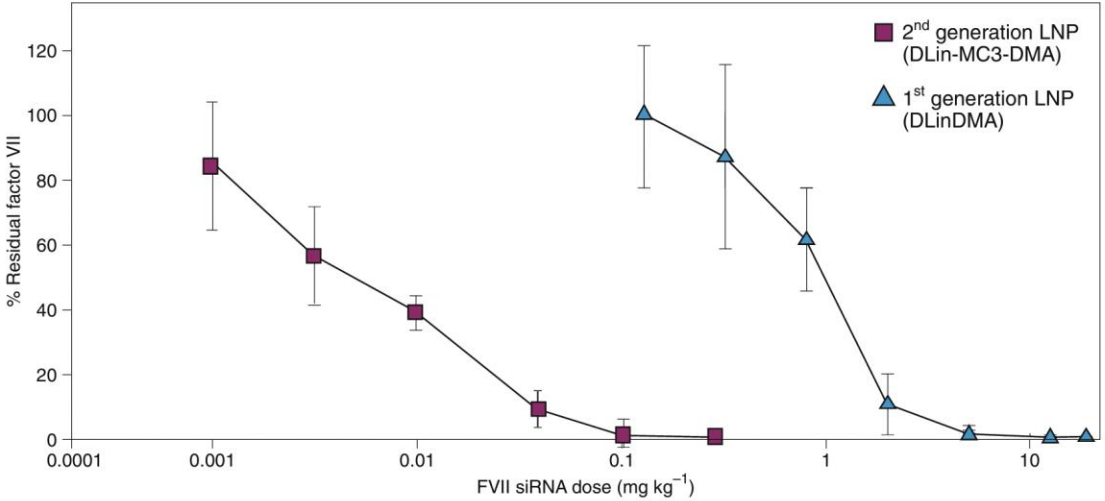
Hereditary transthyretin-mediated amyloidosis (hATTR)

- hATTR is a rare, fatal disease affecting ~50,000 patients worldwide
- hATTR is an inherited condition
- A variant in the TTR gene causes the TTR protein to take on an abnormal shape and misfold
- The misfolded TTR protein forms damaging clumps called amyloid deposits
- TTR protein is primarily produced in the liver



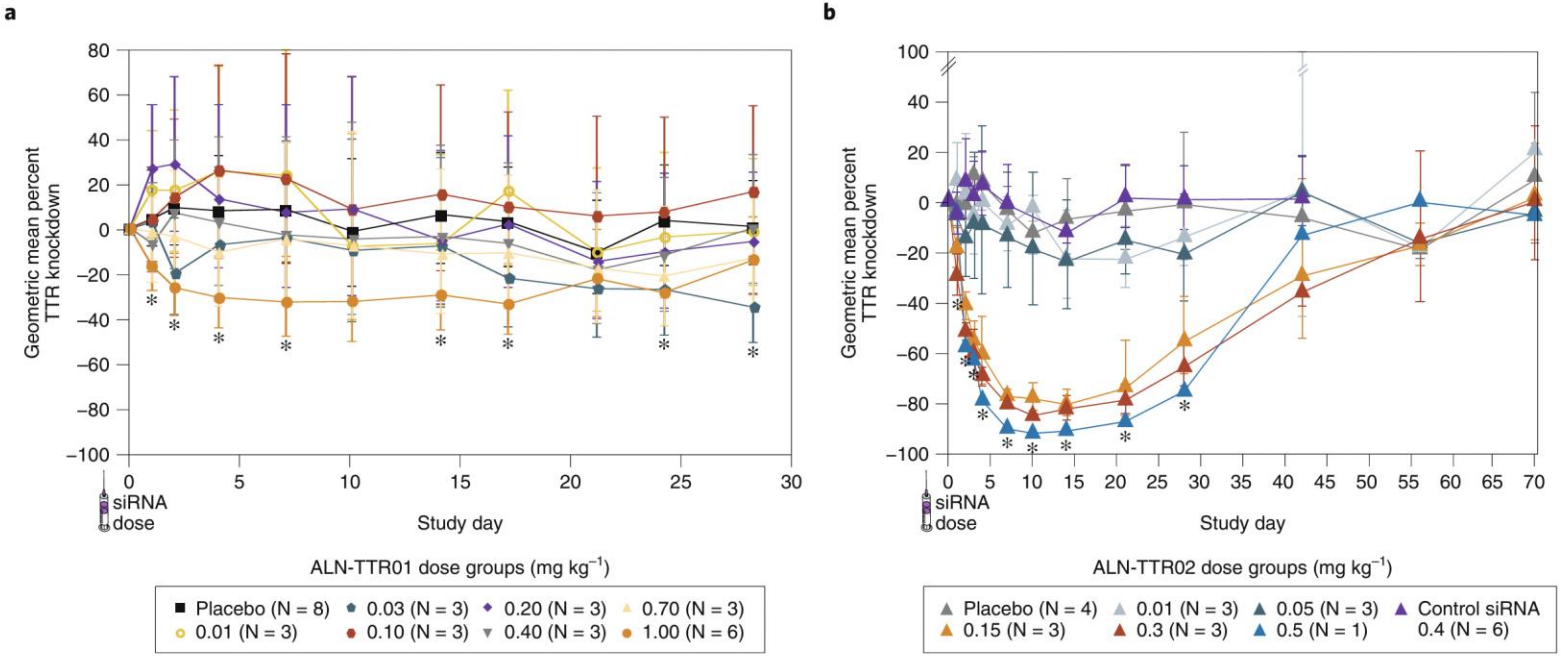
Onpattro (patisiran) – the 1st siRNA-drug

- LNPs have a narrow therapeutic index
- Two Ionizable lipid candidates
  - DLinDMA LNPs
  - MC3-containing LNPs showed a ~100-fold improved potency compared with DLinDMA LNPs



Onpattro (patisiran) – the 1st siRNA-drug

- Clinical testing
  - ALN-TTR01 and ALN-TTR02 contained identical TTR-targeting siRNA
  - Packaged in either DLinDMA LNPs or MC3 LNPs



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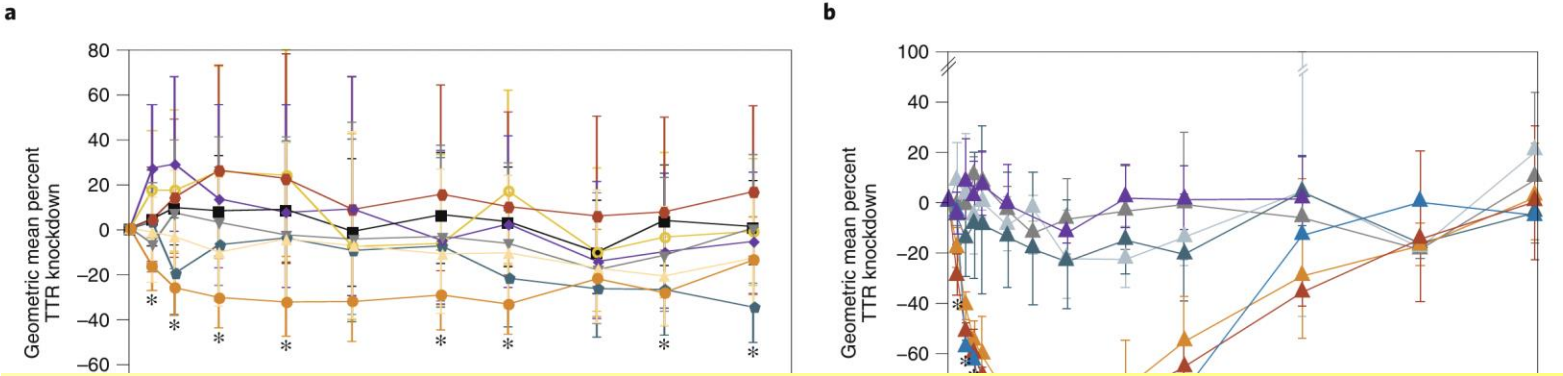


- MC3 LNPs enabled potent TTR silencing and an acceptable tolerability profile
- MC3 LNP ALN-TTR02 ultimately became the first approved RNAi therapeutic, Onpattro (patisiran)

ALN-TTR01 dose groups (mg kg^{-1})

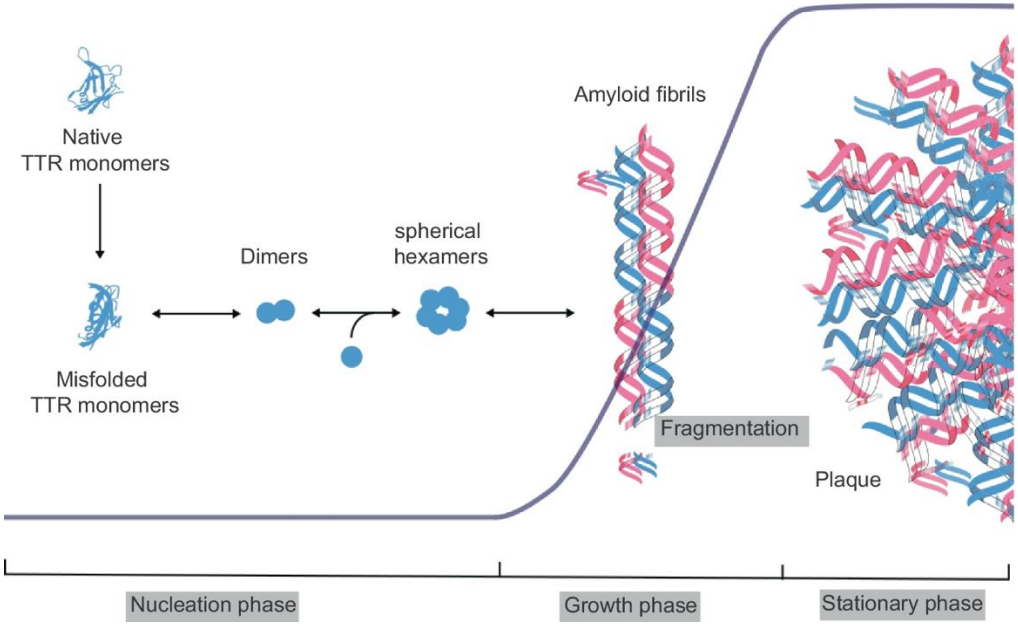
■ Placebo (N = 8) ● 0.03 (N = 3) ◆ 0.20 (N = 3) ▲ 0.70 (N = 3)
○ 0.01 (N = 3) ● 0.10 (N = 3) ▼ 0.40 (N = 3) ● 1.00 (N = 6)

ALN-TTR02 dose groups (mg kg^{-1})

▲ Placebo (N = 4) ▲ 0.01 (N = 3) ▲ 0.05 (N = 3) ▲ Control siRNA 0.4 (N = 6)
▲ 0.15 (N = 3) ▲ 0.3 (N = 3) ▲ 0.5 (N = 1)

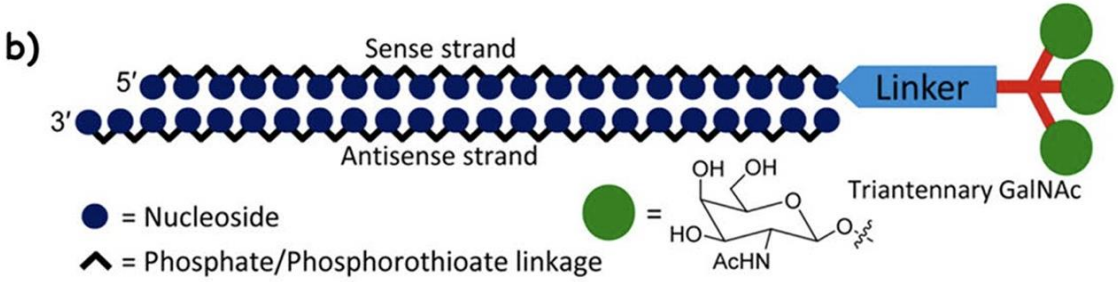
Wildtype transthyretin-mediated amyloidosis (ATTRwt)

- ATTRwt is age related and mainly affects the heart
- ATTRwt affects 200,000 to 300,000 patients worldwide
- TTR protein becomes unstable, misfolds and forms amyloid deposits in various organs of the body
- Alnylam developing siRNA treatment for ATTR amyloidosis with cardiomyopathy (ATTR-CM)



Exploring GalNAc conjugation and backbone stabilization

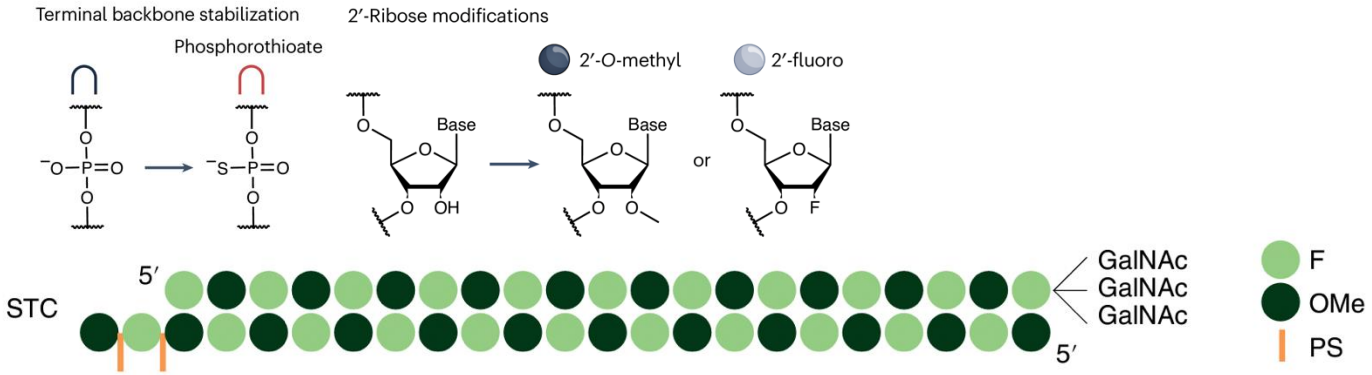
- GalNAc is a ligand for ASGPR
- GalNAc conjugation allow for targeted delivery to the liver
- Alnylam struggled with GalNAc-siRNA potency
- Decide to pursue backbone stabilization
- With initial backbone stabilization, they achieved high potency with lower dose



siRNA–GalNAc Conjugate

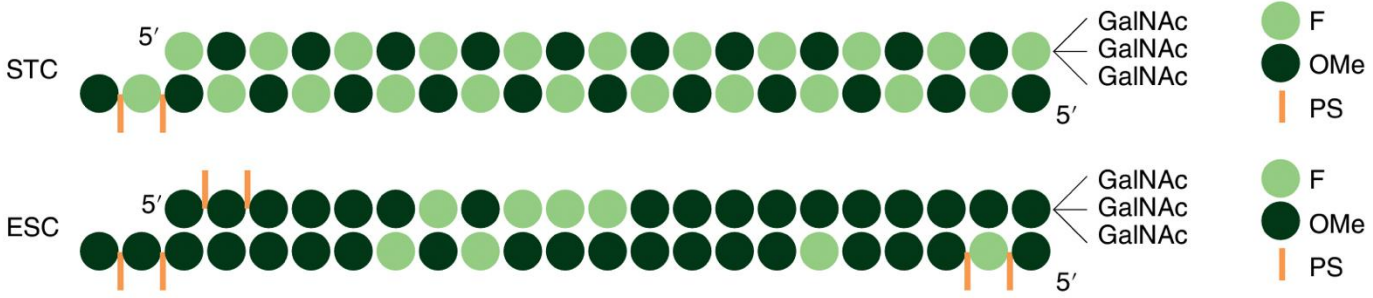
Alnylam chemistry – STC

- Standard Template Chemistry (STC)
  - Alternating 2′-O-methyl and 2′-fluoro modifications are placed throughout both strands
- Required high dose and poor toleration, i.e. Revusiran (discontinued in Phase 3 trial)



Alnylam chemistry – ESC

- Enhanced Stabilization Chemistry (ESC)
  - Contains two additional phosphorothioate (PS) linkages at the 5′ ends of both strands
  - Contains fewer 2′-fluoro substitutions, which further improves stability
- The first clinical program was for fitusiran for the treatment of hemophilia
- However, in early clinical studies, a 10–20% incidence of liver enzyme elevations due to off-target effects



ESC-GalNAc-siRNA conjugates

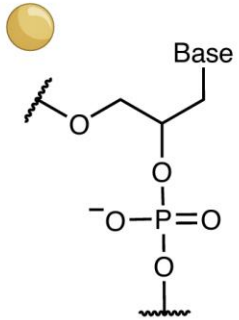
- Givlaari (givosiran), for acute hepatic porphyria (approved 2019)
- Leqvio (inclisiran), for adults with heterozygous familial hypercholesterolemia or clinical atherosclerosis requiring additional lowering of LDL cholesterol (approved 2021)
- fitusiran for hemophilia A or B

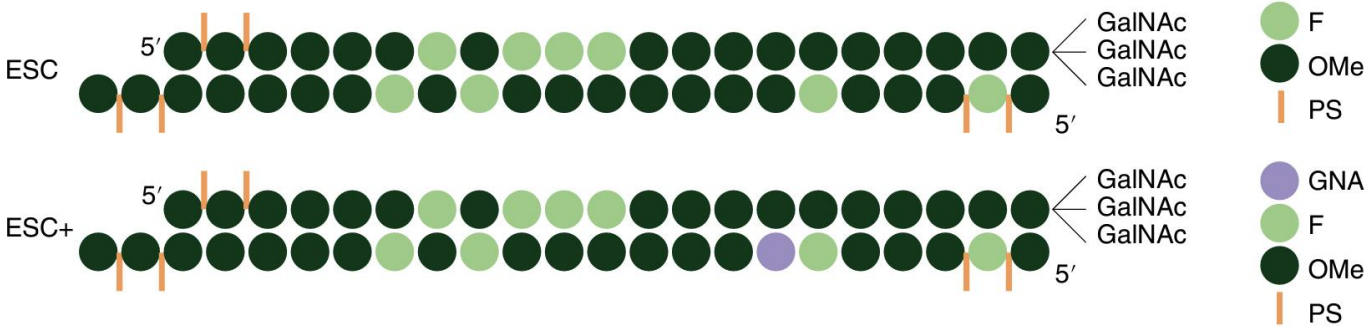
- The ESC-GalNAc-siRNA conjugates
  - Higher potency than STC with a 200-fold lower exposure
  - Unexpected improvement in durability, enable dosing once every 6 months

Alnylam chemistry – ESC+

- A solution was to introduce a destabilizing glycol nucleic acid (GNA) nucleotide into the seed region of the antisense strand
- This weakens the interaction between the siRNA and unintended mRNA
- At position 6 or 7 of the guide strand
- Substantially reduces off-targeting with minimal impact on primary target activity

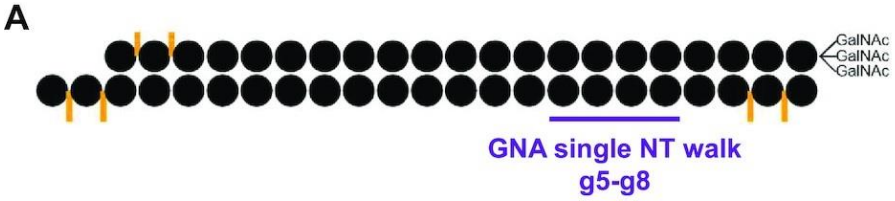
Glycol nucleic acid

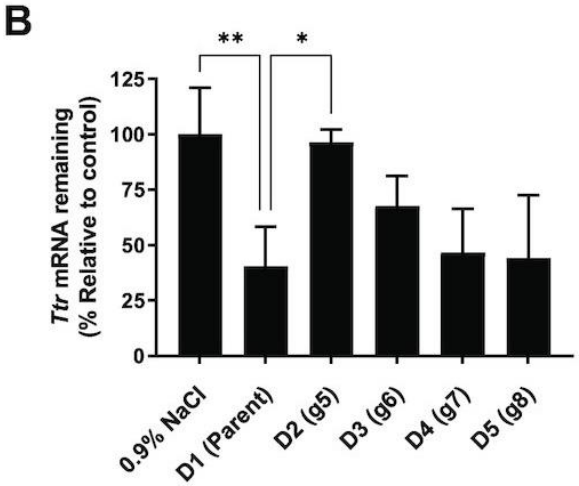




Alnylam chemistry – ESC+

- When placed at position 7 of the antisense strand, GNA efficiently mitigates off-target effects



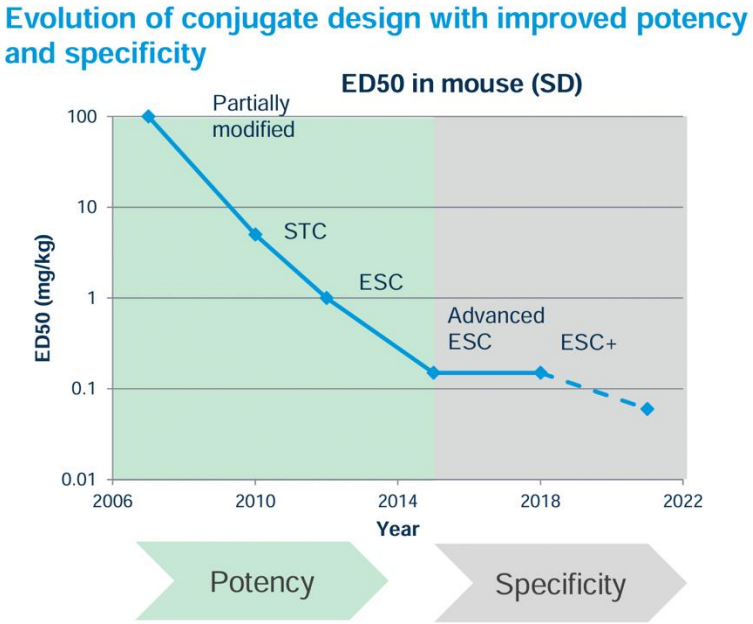


Alnylam chemistry – IKARIA

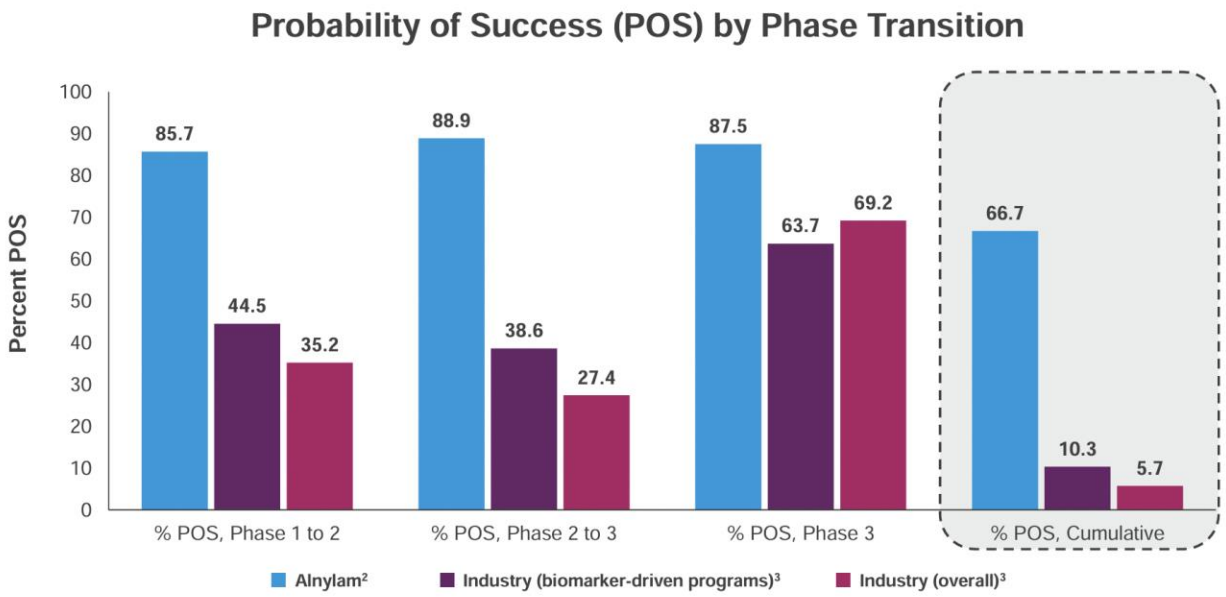
- The latest Alnylam platform
- Used in Nucresiran for treating transthyretin (ATTR) amyloidosis
- Nucresiran is in Phase 1
- Nucresiran demonstrated the potential to achieve durability supportive of biannual or annual dosing

Alnylam chemistry evolution

- STC -> ESC -> ESC+
- Greater potency, higher specificity, and lower toxicity



High-yield of productivity

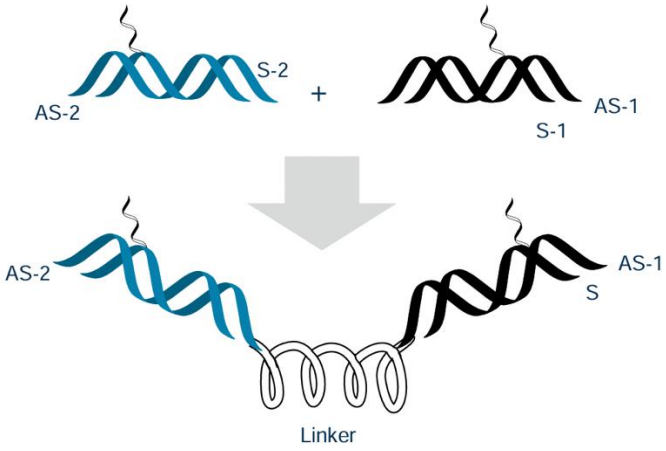


siRNA outlook

Multivalent siRNA

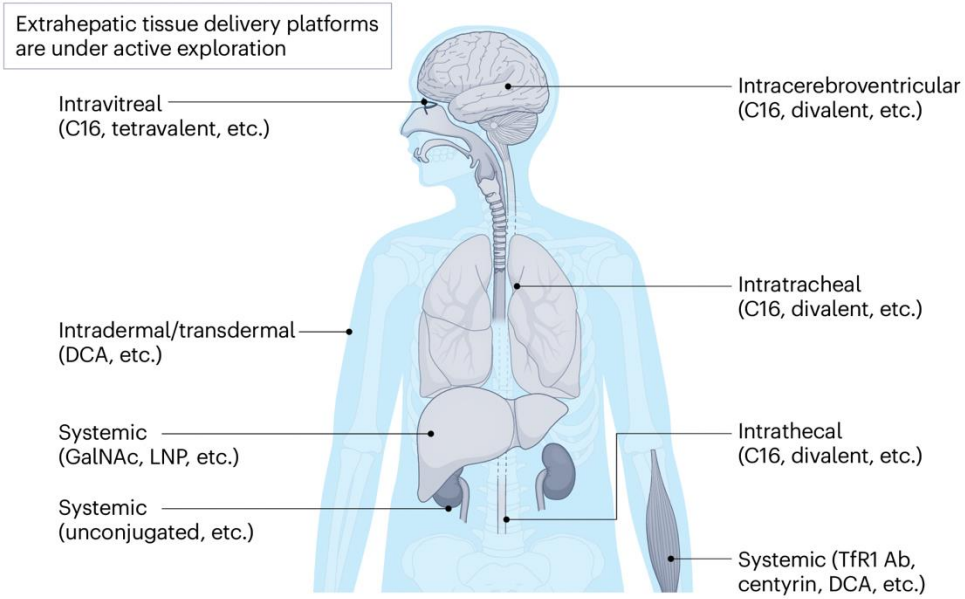
- Multivalent scaffolds
  - Covalently link two or more siRNAs that target the same gene
  - Increase in molecular weight leads to reduced kidney filtration/clearance
  - Showed robust tissue distribution and efficacy in the lung and eyes

- GEMINI platform by Alnylam
  - Simultaneous silencing of two transcripts or same using single chemical entity
  - Under development for targeting the central nervous system (CNS)



Extrahepatic delivery

- Advancements in molecular design and extrahepatic delivery platforms
- Expecting a potential major expansion in the utility of siRNA drugs across various indications



EN 580.133
ASOs as Therapeutics (Part I)

Elaine Zhelan Chen

Jan 13, 2025

Created by Elaine Zhelan Chen

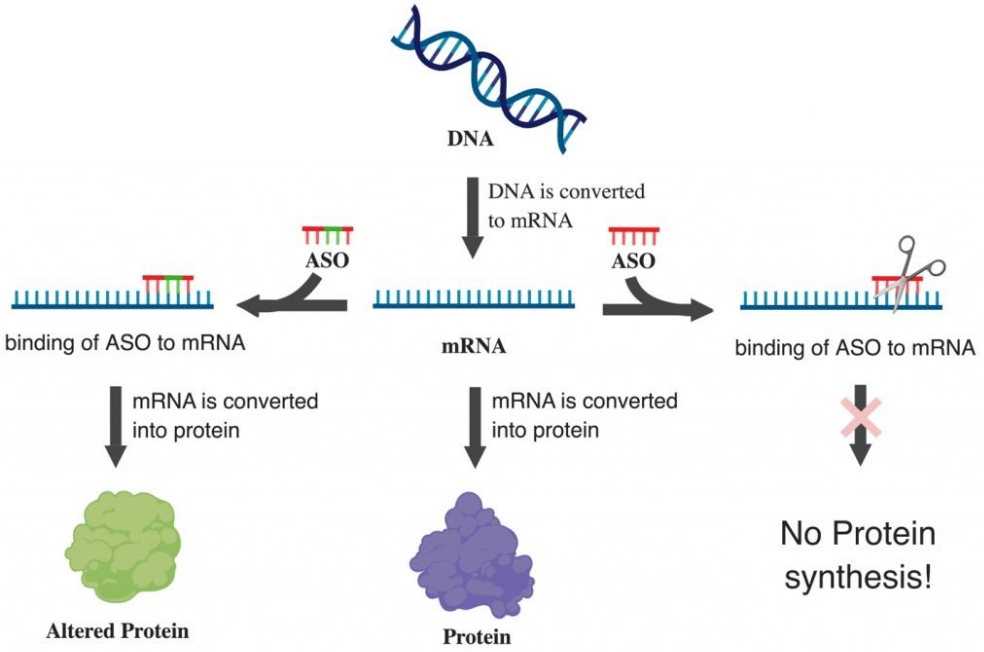
Emergence of RNA therapeutics

Antisense oligo (ASO)

- First application of RNA base-pairing for therapeutic purposes described in 1978
  - Inhibit Rous sarcoma virus (RSV) viral replication using ASO
- Discovery of RNA splicing in 1977
- First study to show that ASO can modulate splicing in 1993
  - Showed therapeutic potential for splicing variation related diseases
- First ASO drug approved by the FDA in 1998
  - Fomivirsen by Ionis Pharmaceuticals

ASO

- Antisense oligonucleotides
- ASOs are short (~18-30 nt), single-stranded oligonucleotides that can bind to RNA in a target-specific manner and ultimately modify protein expressions
- ASOs can be DNA, RNA, or a mix of DNA and RNA



ASO mechanisms

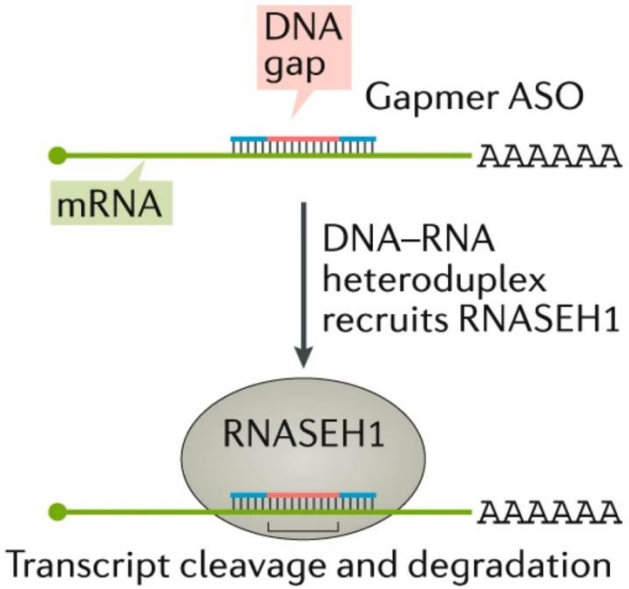
ASO mechanisms

- RNase H
- Steric block
  - modulation of alternative splicing
  - Translation inhibition
  - Increase translation

- RNA interference
  - siRNA is sometimes discussed as a ASO drug

RNase H mediated mRNA cleavage

- DNA-based ASO, also known as Gapmer ASO, binds to target mRNA transcripts
- Endogenous RNase H enzyme RNaseH1 recognizes RNA–DNA heteroduplex
- RNaseH1 cleaves at the site of ASO binding results in destruction of the target RNA and thereby silencing target gene expression
- RNaseH1 is active in both the cytoplasm and the nucleus
- This approach is used to downregulate disease-causing genes
- Approved ASO drugs: fomivirsen, mipomersen and inotersen

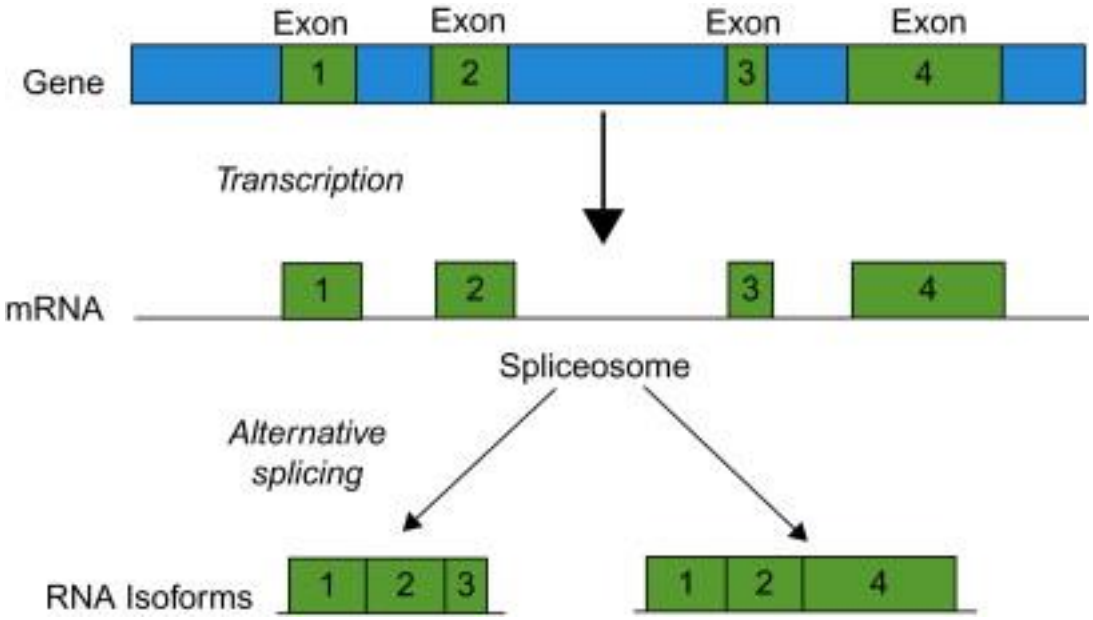


Steric block

- Do not induce target mRNA degradation
  - No consecutive DNA-like bases to avoid recruitment of RNaseH1
- Mask specific sequences within a target transcript to interfere with transcript RNA–RNA and/or RNA–protein interactions
- Can results in
  - Decreased protein level
  - Increased protein level
  - Expression of a protein isoform

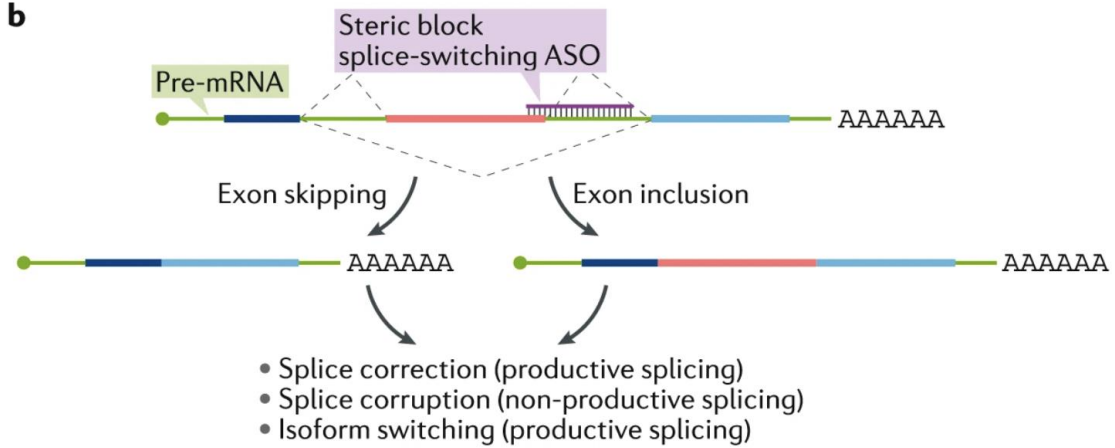
Splicing

- Splicing removes non-coding regions of RNA (introns) and joins coding regions (exons) together to create mature messenger RNA (mRNA)



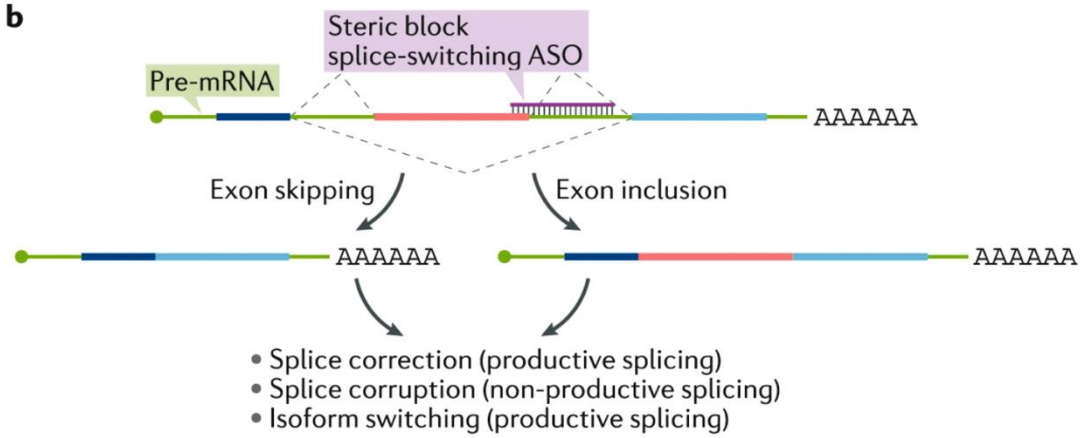
Steric block – alternative splicing

- ASO can modulate alternative splicing to selectively exclude or retain a specific exon(s)
- ASO can mask a splicing signal so that it becomes invisible to the spliceosome, leading to alterations in splicing



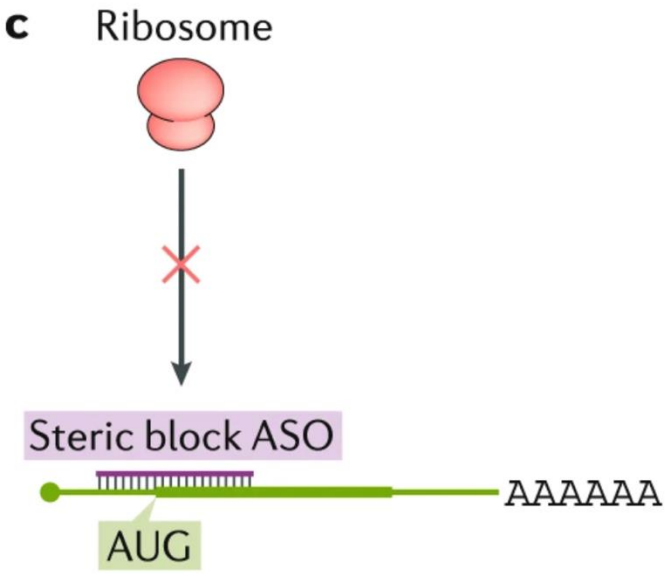
Steric block – alternative splicing

- Splice correction
  - Restore the translational reading frame to rescue production of a therapeutic protein
- Splice corruption
  - Deliberately skip an exon to disrupt the translation of the target gene
- Isoform switching
  - Diminish the expression of harmful protein isoforms and/or promoting the expression of beneficial ones
- Approved ASO-drugs: eteplirsen, golodirsen and nusinersen



Steric block – translation inhibition

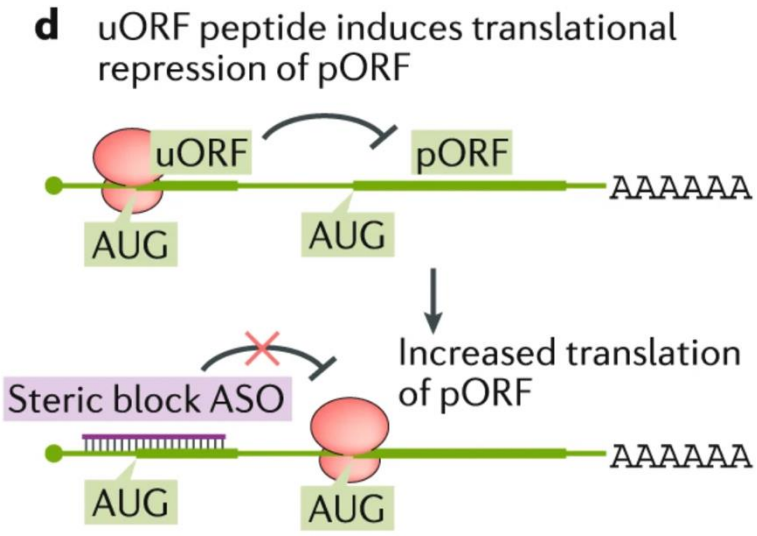
- ASO can prevent the ribosome from translating the RNA, and thus reduce protein level without reducing RNA level
- Target a sequence either in the 5'UTR just upstream of the start codon or or within ~25 bases past the start codon



Steric block – increase translation

- AUG upstream of the canonical AUG gives rise to upstream open reading frames (uORFs)
- The upstream AUG will compete with the canonical AUG for translation initiation by the ribosomes
- ASOs can sterically block translation initiation at the uORFs and promote translation at the canonical ORFs

- A large fraction of human genes have uORFs
- Studies have shown a 30% to 150% increase in protein abundance by blocking uORFs with ASOs



Problem set #3

- **Due Wednesday, Jan 15th, 5PM EST**

Attendance